

Replicating ‘The Role of Time-Varying Price Elasticities in Accounting for Volatility Changes in the Crude Oil Market’ (Baumeister and Peersman, 2013)

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Motivation

- Baumeister and Peersman (2013) provides documented datasets and codes.
- The effort here consisted in translating the MATLAB code to C++ (Armadillo) and R for a fully portable replication.
- I used Vargas Sepúlveda (2025) for a straightforward open source rewrite of the MATLAB code.

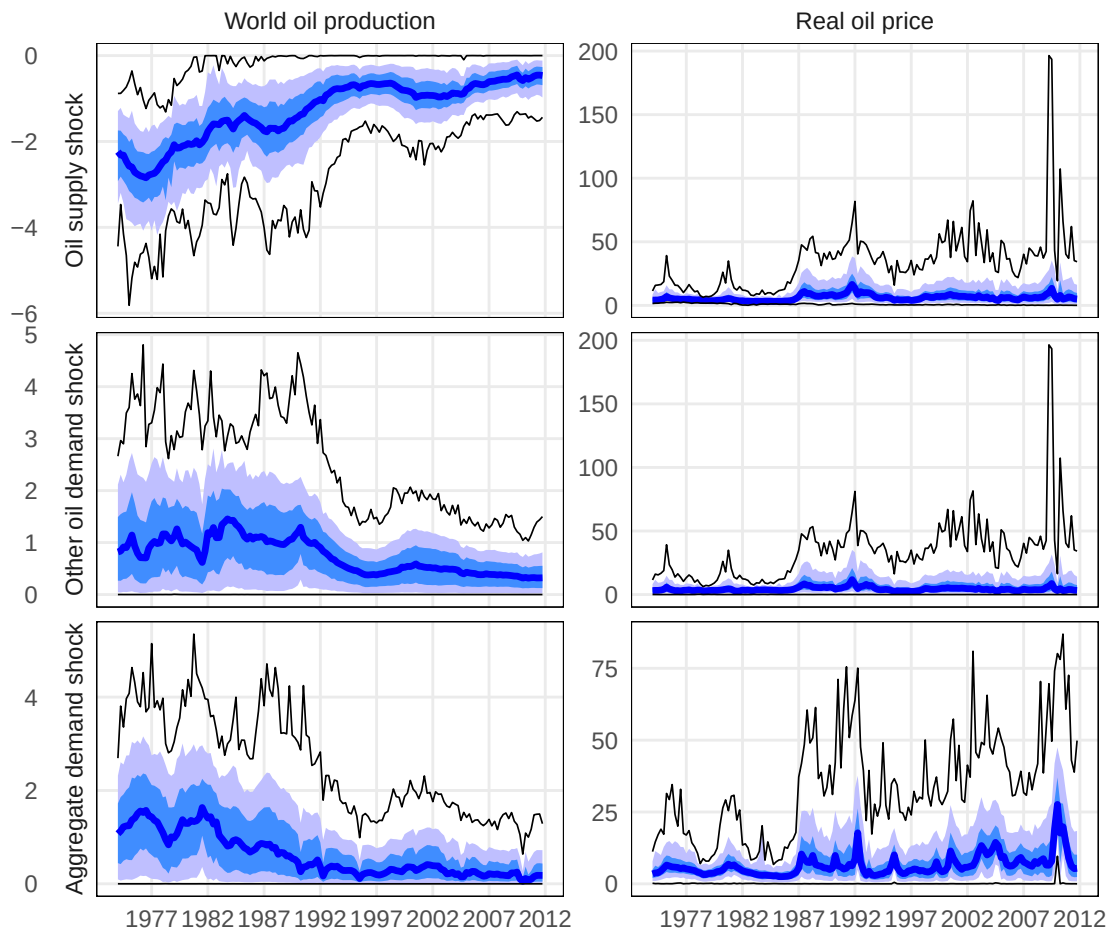
Technique and Data

- Time-varying parameter VAR (TVP-VAR-SV) to model the crude oil market because:
- Suitable for the oil sector where new production in response to price incentives will be gradual and subject to capacity constraints.
- Efforts toward other energy sources are also reflected in slow variation in the energy share over time.
- Burns = 50,000; Draws = 5,000

Code and reorganized data: <https://github.com/pachadotdev/ecod025replication>

Replication of Fig 2. Time-varying median impact impulse responses

- Dark blue lines: Median response
- Shades of blue areas: 68% and 95% posterior credible intervals resp.
- Black lines: Range of admissible models



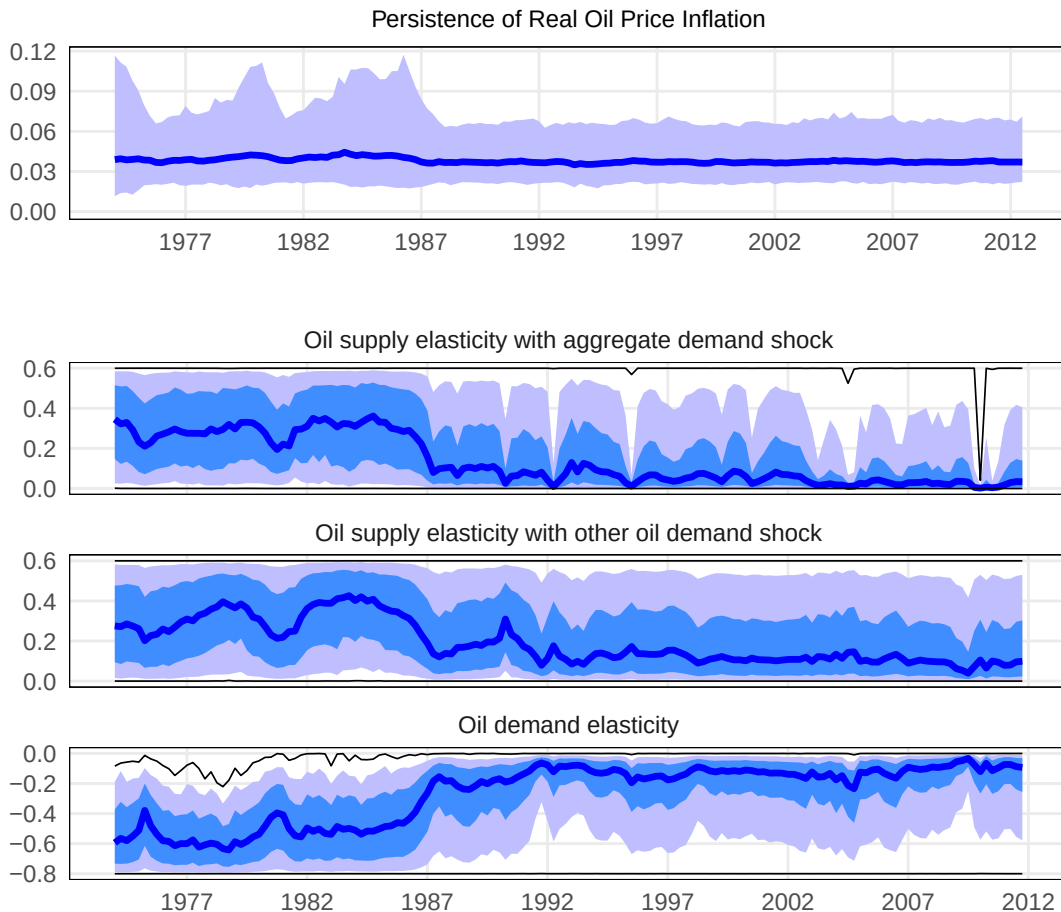
Rep. of Fig 3. Spectrum of price inflation and Fig 4. Short-run price elasticities

For Fig 3 (top panel):

- Dark blue line: Median response
- Shade of blue area: 95% posterior distribution.

For Fig 4 (bottom three panels):

- Dark blue line: Median response
- Shade of blue areas: 68% and 95% posterior credible intervals resp.
- Black lines: Range of admissible models



- Replication effort: Medium but for a good reason (elegant model).
- Results: Identical to the original paper.
- The article and supplementary material allow to match the results, and therefore an extension of this would be to conduct a complete replication.
- The testing model ($B=5,000$; $D=500$) took around 30 minutes to run to then correct and test again until matching the results.
- The full model as in the article ($B=50,000$; $D=500$) took around 2 hours to run to then correct and test again until matching the results.
- One major scalability drawback is the computational cost of Bayesian estimation with MCMC.

References

- Baumeister, Christiane, and Gert Peersman. 2013. "The Role of Time-Varying Price Elasticities in Accounting for Volatility Changes in the Crude Oil Market." *Journal of Applied Econometrics* 28 (7): 1087–1109. <https://doi.org/10.1002/jae.2283>.
- Vargas Sepúlveda, Mauricio. 2025. "Armadillo4R." <https://github.com/pachadotdev/armadillo4r>.